Eric **Zhang Shao Jun**

Availability: April 2026 - August 2026

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SKILLSETS

Skills: Power BI, Tableau, Microsoft Office (Proficient), Autodesk Revit/Inventor
Languages: Python, PostgreSQL, VBA, R (Proficient), Java, JavaScript, HTML, CSS
Libraries/Tools: Pandas, NumPy, PyTorch, Flask (Python), pgAdmin (PSQL)
Modules: Machine Learning, Data Analytics Tools: Data Visualization, Data Structures, Database Management

EDUCATION

National University of Singapore **August 2023 – May 2027**
Bachelors in Industrial Systems Engineering and Management
with Second major in Data Science and Analytics
Minor in Data Engineering
Activities: NUS Varsity Sailing Team, NUSCAS EXCO

AWARDS

Colours Award (Sailing)
Represented Singapore Schools Sports Team1 and achieved top four in individual/ team/ division at the NSG.
CDC/CCC Education Merit Award
Celebrating achievements of students who have done well academically (top 25%)

WORK EXPERIENCE

Systems on Silicon Manufacturing Company (TSMC Joint Venture), *SPC/MSA Internship*
Skillsets: OCR, SQL, Power BI, Tableau, VBA **July 2025 – December 2025**

- Assisted lead MSA Engineer in automating e-calibration processes using Google Vision's OCR technology, achieving calibration pass/fail detection accuracy of 83.6% for printed media, and 92% for printed text.
- Led the monthly reporting of CPK (Process Capability Index) and OOC (Out-of-Control) wafer statistics, automating data analysis through VBA, achieving over 70% efficiency improvement in recurring workflows.
- Spearheaded the migration of the Tool Calibration dashboard from Tableau to Power BI, learning both tools concurrently and improving report accessibility, leading to a 30% increase across all SSMC departments.
- Utilized TSMC's in-built SPC software and Oracle SQL database to identify and remove duplicate SPC charts during part-ID conflicts, ensuring accuracy in wafer process data.

Zhongshi Chemical Engineering, *Internship*
Skillsets: Excel, Powerpoint, AutoCAD **December 2024 – January 2025**

- Logged and managed data by transferring entries from CAD systems to Excel spreadsheets.

Kenyon Pte. Ltd, *Systems Engineering Internship*
Skillsets: VBA, Autodesk Revit/Inventor, Siemens SIMATIC Step-7 **January 2021 – March 2021**

- Worked in the instrumentals department in an engineering company
- Built macro tools to automate the conversion of PLC signal logs and configuration data into standardized instrumentation layout files, reducing manual formatting effort by ~50%.
- Learnt practical skills, developed a proficient skill in using AutoCAD software, such as Revit and Inventor

RELEVANT PROJECTS

ML: Wafer Defect Classification with Deep Learning (PyTorch)

- Goal: Correctly identify multi-class wafer defects given a 52x52 wafer map.
- Engineered a multi-layer CNN featuring convolutional "scanners" for feature extraction.
- Implemented a robust training pipeline using Cross-Entropy Loss and the Adam optimizer, achieving high classification accuracy across 8 distinct defect categories (e.g., Edge-Ring, Donut, Center).
- Achieved a final model accuracy of 98.1% using unseen test data.

ML: Household Electricity Consumption Forecasting with Time Series (Keras)

- Goal: Predict hourly electricity consumption 24 hours into the future using one week of historical data.
- Engineered three distinct models (ARIMA, CNN, and LSTM) incorporating cyclical encoding (sine/cosine transformations) to capture recurring daily and weekly seasonal patterns.
- Evaluated performance using MSE and MAE, discovering that LSTM performed the best with MAE 0.43

Relational Database Design & Management with PostgreSQL (pgAdmin)

- Goal: Developed a relational database for car rental operations, ensuring data integrity and querying.
- Designed an ERD and normalized tables (3NF) for customers, cars, rentals, and passengers, enforcing constraints like unique NRIC and license plate format validation
- Stored Procedures & Business Logic: Developed PSQL functions and triggers to enforce rules (e.g., ensuring at least one licensed driver per rental) and automated record-keeping for historical analysis.